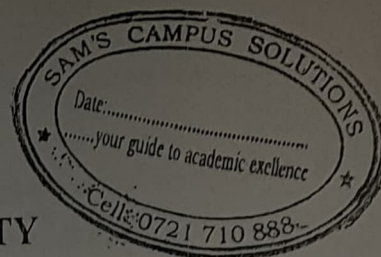


ECU 100



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017

**FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE, CIVIL, MECHANICAL, ELECTRICAL, BIOSYSTEMS, BIOMEDICAL,
AEROSPACE, PETROLEUM**

ECU 100: CHEMISTRY FOR ENGINEERS 1

DATE: Monday, 5th December 2016

TIME: 2.00 p.m. - 4.00 p.m.

INSTRUCTIONS:

1. Question ONE is **COMPULSORY**
2. Answer question ONE and any two other questions

Useful information:

Periodic Table provided

Plank's constant $h=6.6262 \times 10^{-34} \text{ Js}$

Speed of light $=3 \times 10^8 \text{ m/s}$

Rydberg's constant; $R=109677.8 \text{ cm}^{-1}$

Bohr's constant; $B=2.179 \times 10^{-18} \text{ J}$

Avogadro's Number; $N_A=6.022 \times 10^{23} \text{ photons}$



QUESTION ONE (20 marks)

- a) Discuss briefly the information given by the subsidiary quantum number (l).

(6 marks)

- b) List FOUR parameters that describe an electromagnetic wave

(4 marks)

- c) In which group of the periodic table will be found:

- (i) An element with atomic number 20

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

(ii) An element having $2s^2 2p^2$

(2 marks)

d) Write the electronic configuration of the elements whose atomic numbers are 24, 58, 14 and 46.

Use the configuration to identify their group and period in the periodic table.

(8 marks)

QUESTION TWO (25 marks)

a) An atom has 2K, 8L and 2M electrons. Find the total number of:

(i) Protons

(ii) (ii) sub-shells

(iii) p-electrons

(iv) shells

(v) paired electrons

(vi) principal quantum number

(6 marks)

b) The electronic configuration of Carbon is $1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^0$ and not $1s^2, 2s^2, 2p_y^0, 2p_z^0$. Explain

(3 marks)

c) Write correct orbital notation for each of the following set of quantum numbers:

(i) $n=1, l=0, m=0$

(ii) $n=2, l=1, m=1$

(iii) $n=3, l=2, m=-1$

(3 marks)

d) Write a note on Aufbau's principle.

(5 marks)

e) Discuss FOUR general characteristics of Periods of the Periodic table

(8 marks)

QUESTIONS THREE (25 marks)

a) Draw the Lewis structure of the following molecules and explain their formation:

(i) NH_3

(ii) Al_2O_3

(iii) BF_3

(6 marks)

b) Give FIVE properties of Ionic Compounds

(5marks)

c) (i) For the principal quantum number $n=3$, what are the possible values for l , the angular quantum number.

(2 marks)

(ii) If the Angular quantum number $l=3$ what are the possible values for the magnetic quantum number m

(2 marks)

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- 6) How many nodes do the following orbitals possess
- 4p
 - 3s

(2 marks)

- c) Write the electronic configuration of the following species

- | | | |
|--------|----------------------|---------|
| (i) Cr | (ii) S ²⁺ | (iii) I |
| (iv) V | (v) Ni | (vi) Ar |

(6 marks)

- d) Write the structure of the compound whose IUPAC name is 5-ethyl-2,2-dimethylheptane

(2 marks)

QUESTION FOUR (25 marks)

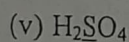
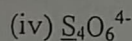
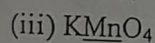
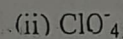
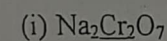
- a) Differentiate between ionic and covalent compounds

(8 marks)

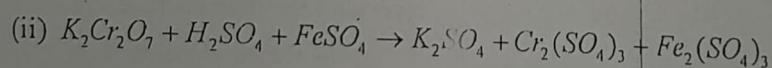
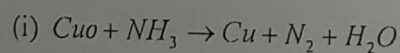
- b) Outline the rules for calculating oxidation numbers

(3 marks)

- c) Indicate the oxidation number of the underlined atoms in each of the following:



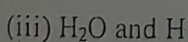
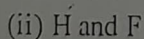
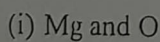
- c) Balance the following equations by oxidation number method



(10 marks)

QUESTION FIVE (25 marks)

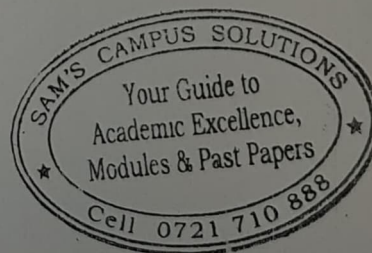
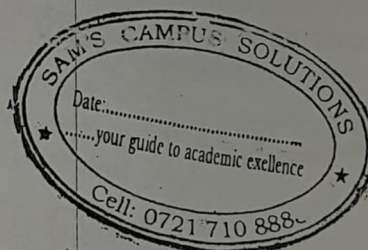
- a) Indicate the type of bonds formed by these elements



(3 marks)

- b) List FOUR properties of Organic compounds and differentiate them from Inorganic compounds

(4 marks)



c) What is electromagnetic radiation?

(2 marks)

d) Calculate the wave number and the frequency of yellow light having a wave length of 650nm

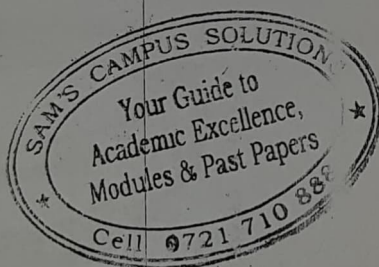
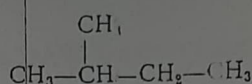
(5 marks)

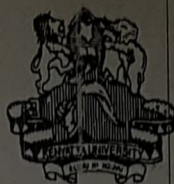
e) Two lines one violet and the other red, in a spectrum have wavelengths equal to 4000Å and 6000Å respectively. Calculate the energies of photons transmitted by them

(8marks)

f) Name the following structure. Say why the prefix number is not needed.

(3 marks)





KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE

ECU 100: CHEMISTRY FOR ENGINEERS I

DATE: Monday 19th February 2018

TIME: 8.00a.m -10.00a.m

INSTRUCTIONS:

1. Question ONE is COMPULSORY
2. Answer question ONE and any two other questions

Useful information:

Periodic Table provided

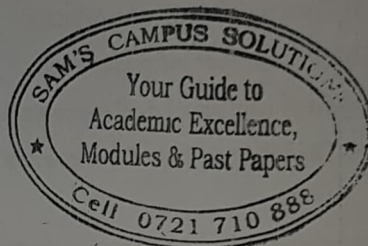
Plank's constant $h=6.6262 \times 10^{-34}$ Js

Speed of light $=3 \times 10^8$ m/s

Rydberg's constant; $R=109677.8 \text{ cm}^{-1}$

Bohr's constant; $B=2.179 \times 10^{-18}$ J

Avogadro,s Number; $N_A=6.022 \times 10^{23}$ photons



QUESTION ONE (20 marks)

a) Describe briefly the following:

(i) Thomson's atomic Model

(3 marks)

(ii) Rutherford's Nuclear Atomic Model

(3 marks)

b) Give a definition of the following terms:

(i) Wave number

(ii) Frequency

(iii) Spectrum

(iv) Oxidation number

(v) Atomic orbital

(5 marks)

c) There are two elements having atomic number 7 and 9

(i) Write their electronic configuration

(ii) Which has maximum number of unpaired electrons?

(iii) What is the valency of each?

(4 marks)

d) List FIVE differences between oxidation number and valency.

(5 marks)

QUESTION TWO (25 marks)

a) A neutral atom of an element has two K, eight L and five M electrons. Give the following information:

(i) Atomic number

(ii) Total number of s-orbitals

(iii) Total number of p-electrons

(iv) Total number of d-electrons

(v) The total number of f-electrons

(7 marks)

b) How many orbits, orbitals and electrons are there in an atom having atomic weight 23 and atomic number 11?

(5 marks)

c) Write the electronic configuration (with electron spins) of the following:

(i) Elements with atomic number 7, 21, 4

(4 marks)

(ii) Ions Fe^{2+} , Br^-

(5 marks)

d) Write a note on Aufbau's principle.

(5 marks)

e) Discuss FOUR general characteristics of groups of the Periodic table

(4 marks)

QUESTIONS THREE (25 marks)

a) Draw the Lewis structure of the following molecules and explain their formation:

(i) PCl_3

(ii) MgO

(iii) NH_3

(6 marks)

b) Outline FIVE properties of Covalent Compounds

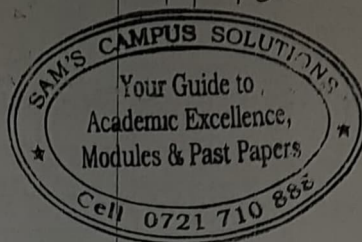
(5 marks)

ECU 100

19th Feb 2018

c) Define the following:

- (i) The Octet rule
- (ii) Pauli's Exclusion Principle
- (iii) Co-ordinate Bond
- (iv) Electronegativity



(6 marks)

b) Use Lewis dot symbols to show the formation of aluminum oxide (Al_2O_3)

(4 marks)

c) Give the steps used in writing the Lewis structures of compounds

(4 marks)

QUESTION FOUR (25 marks)

a) Discuss briefly the factors that favor the formation of Ionic compounds

(10 marks)

b) Outline the rules for calculating oxidation numbers

(3 marks)

c) Calculate the oxidation number of:

(i) Cr in $Cr_2O_7^{2-}$ and CrO_2Cl_2

(ii) Mn in MnO_4^- and MnO_4^{2-}

(iii) S in H_2SO_4

(10 marks)

c) Differentiate between Oxidation and Reduction reactions

(2 marks)

QUESTION FIVE (25 marks)

a) Name the Scientist and with the aid of a diagram, describe the experiment used to determine photo electric effect,

(4 marks)

b) Organic compounds are abundant in nature and have a very wide application in industry. Outline the properties of these compounds.

(6 marks)

c) Determine the frequency of radiation emitted by the transition of an electron in the 2nd of the Paschen series.

(4 marks)

d) What is the wave length of light emitted when hydrogen atom undergoes a transition from level $n=4$ to level $n=2$?

(5 marks)

e) Calculate the wave number and frequency of yellow light having a wave length of 650nm

(6 marks)

Periodic Table of the Elements

ECU. 100.

Main-group elements

- ☐ Metals
- ☐ Nonmetals
- ☐ Metalloids
- ☐ Noble gases

Transition elements

Main-group elements

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1A	2A	3A	4A	5A	6A	7A	8A	9A	10A	11A	12A	13A	14A	15A	16A	17A	18A
1 H 1.00794	2 He 4.00260	3 Li 6.941	4 Be 9.01218	5 B 10.811	6 C 12.011	7 N 14.0067	8 O 15.9994	9 F 18.9984	10 Ne 20.1797	11 Na 22.9898	12 Mg 24.3050	13 Al 26.9815	14 Si 28.0855	15 P 30.9738	16 S 32.066	17 Cl 35.4527	18 Ar 39.948
19 K 39.0983	20 Ca 40.078	21 Sc 44.9559	22 Ti 47.88	23 V 50.9415	24 Cr 51.9961	25 Mn 54.938	26 Fe 55.847	27 Co 58.9332	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.9216	34 Se 78.96	35 Br 79.904	36 Kr 83.80
37 Rb 85.4678	38 Sr 87.62	39 Y 88.9059	40 Zr 91.224	41 Nb 92.9064	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.906	46 Pd 106.42	47 Ag 107.868	48 Cd 112.411	49 In 114.818	50 Sn 118.710	51 Sb 121.76	52 Te 127.60	53 I 126.904	54 Xe 131.29
55 Cs 132.905	56 Ba 137.327	57 La 138.906	58 Ce 140.115	59 Pr 140.908	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.965	64 Gd 157.25	65 Tb 158.925	66 Dy 162.50	67 Ho 164.930	68 Er 167.26	69 Tm 168.934	70 Yb 173.04	71 Lu 174.967	
87 Fr (223)	88 Ra (226)	89 Ac (227)	90 Th (232)	91 Pa (231)	92 U (238)	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (260)	

☐Metals

☐Nonmetals

☐Metalloids

☐Noble gases

Transition elements

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
5	B	6	C	7	N	O	8	F	9	10	Ne	13	Al	14	Si	15	Ar
10.811	12.011	14.0067	15.9994	18.9984	20.1797	22.9898	24.3050	26.9815	28.0855	30.9738	32.066	35.4527	39.948				

* Lanthanide series	58	59	60	61	62	63	64	65	66	67	68	69	70	71
	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
	140.115	140.908	144.24	(145)	150.36	151.965	157.25	158.925	162.50	164.930	167.26	168.934	173.04	174.967
f Actinide series	90	91	92	93	94	95	96	97	98	99	100	101	102	103
	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
	232.038	231.036	238.029	237.048	(244)	(243)	(247)	(247)	(251)	(252)	(257)	(258)	(259)	(260)

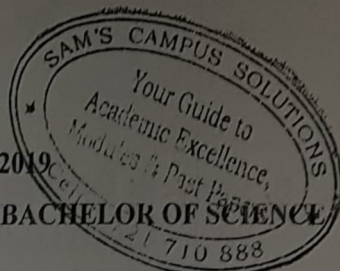
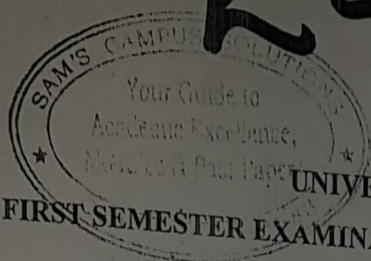
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ECU 100



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2018/2019
FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

ECU 100: CHEMISTRY FOR ENGINEERS I

DATE: Monday, 10th December 2018

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS:

1. Question ONE is COMPULSORY
2. Answer question ONE and any two other questions

Useful information:

Speed of light = 3×10^8 m/s

Plank's constant $h = 6.6262 \times 10^{-34}$ Js

Rydberg's constant; $R = 109677.8 \text{ cm}^{-1}$

Avogadro's Number; $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$

Periodic table is provided

QUESTION ONE (30 marks)

a) Explain the following terms:

- | | |
|-------------------------|---------|
| (i) Wave length | (1mark) |
| (ii) Atomic Orbital | (1mark) |
| (iii) Electronegativity | (1mark) |
| (iv) Covalency | (1mark) |
| (v) Redox reaction | (1mark) |

b) With the aid of an equation, state Plank's quantum theory (2marks)

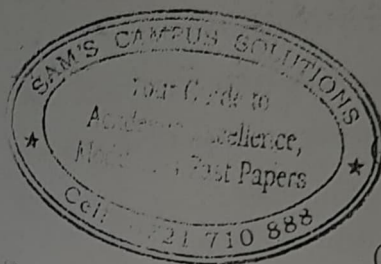
c) The coordinates of an electron can be described by FOUR quantum numbers, state these quantum numbers and outline the information given by each (8marks)

d) Write the electronic configuration of the following species

- | | |
|------------------------|----------|
| (i) Cu | $Z = 29$ |
| (ii) O^{2-} | $Z = 20$ |
| (iii) Fe^{2+} | $Z = 26$ |
| (iv) Ar | $Z = 18$ |

(4 marks)

- 001 123
- e) Draw the Lewis structure for the following compounds: (2 marks)
- (i) CH_4
- (ii) MgO (2 marks)
- f) Identify the homologous series that each of the compounds belong to:
- i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- ii) $\text{CH}_3\text{CH}_2\text{COOH}$
- iii) $\text{CH}_3\text{CH}_2\text{COCH}_3$
- iv) $\text{CH}_3\text{CH}(\text{NH}_2)\text{CH}_2\text{CH}_3$
- v) $\text{CH}_3\text{CHCHCH}_3$
- vi) $\text{CH}_3\text{COOCH}_3$
- vii) CH_3CHO



QUESTION TWO (20 marks)

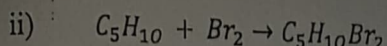
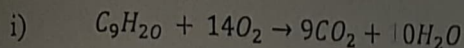
- a) List FOUR key features of the IUPAC periodic table (2 marks)
- b) With the aid of diagrams describe briefly:
- i) Dalton's atomic model (3 marks)
- ii) Rutherford's Nuclear atomic model (3 marks)
- c) Define the following:
- i) Hund's rule of maximum multiplicity (2 marks)
- ii) Rule of half-filled or completely filled orbitals (2 marks)
- d) An atom has atomic number 82
- i) Write its electronic configuration (2 marks)
- ii) To which period and group does it belong? (2 marks)
- e) Describe the following general characteristics of groups in the periodic table:
- (i) Metallic Character (2 marks)
- (ii) Number of electron shells (2 marks)

QUESTIONS THREE (20 marks)

- a) Using Lewis structure describe failure of the octet rule. Give an example in each case. (5 marks)
- b) Outline THREE properties of covalent compounds (3 marks)
- c) Explain any FOUR postulates of Bohr's atomic Theory (6 marks)

d) Complete the following reactions:

(2 marks)



e) Give the values of all the quantum numbers associated with:

(i) the 3p orbitals

(2 marks)

(ii) 5f orbitals

(2 marks)

QUESTION FOUR (20 marks)

a) State the octet theory of valency and outline FOUR main points that the theory emphasizes.

(5 marks)

b) Describe Thomson model of an atom. (In your discussion include, how it described atomic structure and why it was discarded.)

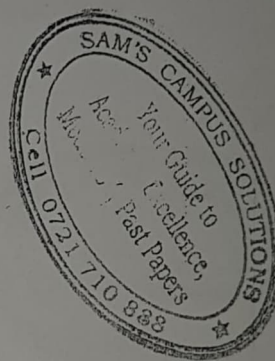
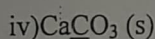
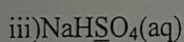
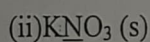
(3 marks)

c) List any FOUR rules applied when calculating oxidation numbers

(4 marks)

d) Calculate the oxidation number of the underlined atoms:

(8 marks)



QUESTION FIVE (20 marks)

a) Indicate the type of bond formed by these elements:

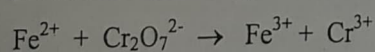
i) Mg and O

(1 mark)

ii) H and F

(1 mark)

b) Balance the following redox reactions in acidic media



(3 marks)

c) Determine the frequency of radiation emitted by the transition of an electron in the 3rd line of the Paschen series.

(4 marks)

d) Calculate the magnitude of the energy of the photon associated with light of wavelength

6057.8 Å

(5 marks)

e) 25 cm³ of 0.5 M HCl was titrated against 0.3 M Na₂CO₃.

(1 mark)

(i) Write the equation for the reaction

(1 mark)

(ii) Calculate the number of moles of HCl used

(1 mark)

(iii) Determine the volume of Na₂CO₃ used.

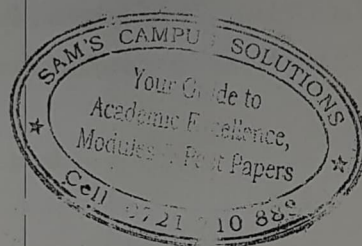
(3 marks)

f) Draw the structural formulas of each of the following compounds

i. 2,2,3-trimethylheptane

ii. 2-methylpent-2-ene

iii. 2-methylbutan-2-ol



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CONCLUSION

Net torque = 0

$\sum \tau = F \Delta r \cos \theta = F \Delta r \sin$

KENYATTA UNIVERSITY

ECU 100 C.A.T 1 January 2021

Avogadro's constant = $6.022 \times 10^{23} \text{ mol}^{-1}$

1. An electron is in 4f orbital. What possible values for the quantum number of n , l and m can it have. (4 marks)
2. What is the maximum number of electrons in
 - i) All 4d orbitals (2 marks)
 - ii) All orbitals with $l=1$ (2 marks)
3. What is the frequency and wave number associated with light of wavelength equal to $8000\text{\AA}$? (7 marks)
 $\lambda = 8000 \times 10^{-10} \text{ m}$ $\nu = 1/\lambda = 1.25 \times 10^{10} \text{ m}^{-1}$
4. Calculate the energy in KJ/mole associated with a radio wave of wavelength equal to $3 \times 10^{13} \text{ m}$. ($h=6.6256 \times 10^{-34} \text{ J.s}$, $c=3 \times 10^8 \text{ ms}^{-1}$) (7 marks)
 $E = hf$
5. (In C.G.S units) Calculate the kinetic energy of an electron emitted from a surface of potassium metal by light of wave length $5.5 \times 10^{-8} \text{ cm}$ (work function $= 3.62 \times 10^{-12} \text{ erg}$). (6 marks)
6. Calculate the wave number and energy of the spectral line in the hydrogen spectrum corresponding to $n_1=3$ and $n_2=5$. The value of Rydberg's constant (R)= $10.97 \times 10^6 \text{ m}^{-1}$. (7 marks)
7. Electromagnetic radiation of wave length 242 nm is just sufficient to ionize the outermost electron of the sodium atom. Calculate the ionization energy of sodium atom ($c=3 \times 10^8 \text{ ms}^{-1}$, $h=6.625 \times 10^{-34} \text{ J.s}$) (6 marks)
 $\lambda = 242 \times 10^{-9} \text{ m}$
8. The wave length of a certain wave is 6 m and 30 waves pass a fixed point per minute. What is the speed of the waves $\lambda = \lambda f$ (3 marks)

1. How many orbits, orbitals and electrons are there in an atom having weight 23 and atomic number 11?
2. Write the full electronic configuration (with electron spins) of the following:
 - (i) Elements with atomic number 8, 21, 42
 - (ii) Ions V^{3+} , Br^- , S^{2-} , Fe^{2+}
3. Arrange the following in the increasing order of oxidation number:
 - (i) Iodine in I_2 , HI , HIO_4 , ICl
 - (ii) Manganese in $MnCl_2$, MnO_2 , $KMnO_4$
4. Which of the reactants act as an oxidizing agent and which is a reducing agent? Explain

$$2H^+ + Mg \rightarrow Mg^{2+} + H_2$$
5. In which group of the periodic table will be found:
 - (i) An element with atomic number 20
 - (ii) An element having $2s^2 2p^2$ outer configuration
 - (iii) An element with three unpaired p-electrons in the valence-shell
6. Draw the Lewis structure of the following molecules and explain their formation:
 - (i) IF_7
 - (ii) $NaCl$
 - (iii) SF_6
 - (iv) OSF_8
7. Use Lewis dot symbols to show the formation of
 - (i) aluminum oxide (Al_2O_3)
 - (ii) Al_2Cl_3
8. Identify the homologous series that each of the compounds belong to and name them:

i) $CH_3CH_2CH_2OH$	ii) CH_3CH_2COOH
iii) $CH_3CH_2COCH_3$	iv) $CH_3CH(NH_2)CH_2CH_3$
v) $CH_3CHCHCH_3$	vi) CH_3COOCH_3
vii) CH_3CHO	
9. Balance the following redox reactions in acidic media
 1. $Fe^{2+} + Cr_2O_7^{2-} \rightarrow Fe^{3+} + Cr^{3+}$
10. Balance the following redox reaction by oxidation number method
 1. $K_2Cr_2O_7 + H_2SO_4 + FeSO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + Fe_2(SO_4)_3$



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2020/2021

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(CIVIL, ELECTRICAL, BIOSYSTEMS, BIOMEDICAL, AEROSPACE, PETROLEUM,
MECHANICAL ENGINEERING)

ECU 100: CHEMISTRY FOR ENGINEERS I

DATE: WEDNESDAY 10TH MARCH 2021

TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS:

1. Question ONE is COMPULSORY

2. Answer question ONE and any two other questions

Useful information:

Speed of light = $3 \times 10^8 \text{ m/s}$

Plank's constant $h = 6.6262 \times 10^{-34} \text{ Js}$

Rydberg's constant; $R = 109677.8 \text{ cm}^{-1}$

Avogadro,s Number; $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$

Periodic table is provided

QUESTION ONE (30 marks)

a) Describe the atomic structure according to the following scientists:

- (i) Neil Bohr (2 marks)
- (ii) J.J. Thomson ✓ (2 marks)
- (iii) Rutherford ✓ (2 marks)
- (iv) Chadwick (2 marks)

b) Explain the following postulates of Bohr's atomic model:

- (i) Fixed circular orbits (2 marks)
- (ii) Stationary energy levels (2 marks)
- (iii) Principle of quantization of angular momentum of the moving electron (2 marks)

c) Enumerate the information provided by the following numbers:

- (i) Principal quantum number (2 marks)

(2 marks) ✓

(ii) Quantum number denoted by symbol l

d) Write the electronic configuration of the following species

(4 marks) ✓

(i) Mn^{2+}

(ii) As^{3+}

(iii) K^+

(iv) Mo

e) Draw the Lewis structure for the following compounds:

(2 marks) ✓

(i) IF_7

(2 marks)

(ii) OsF_8

(2 marks) ✓

(iii) BF_3

(2 marks) ✓

(iv) CH_4

QUESTION TWO (20 marks)

a) Below are general formulas of common organic compounds. Name these compounds and give their functional groups

(4 marks)

(i) $R-C \equiv C-H$

(ii) $R-NH_2$

b) Explain the following giving appropriate reasons:

(2 marks)

(i) Half-filled and completely-filled orbitals are more stable

(2 marks)

(ii) Fe^{2+} ion is less stable than Fe^{3+} ion

c) In which group of the periodic table will be found:

(1 mark)

(i) An element with atomic number 20

(1 mark)

(ii) An element having $2s^2 2p^2$ outer configuration

(1 mark)

(iv) An element with three unpaired p-electrons in the valence-shell

d) Write the electronic configuration of the elements whose atomic numbers are 24, 29, 46, 56 and 78

(5 marks)

e) Explain any two applications of Van der Waals forces

(4 marks)

QUESTIONS THREE (20 marks)

a) Distinguish between covalent and electrovalent bonds

(5 marks)

b) Describe three defects of Mendeleev's periodic table

(3 marks)

c) Find the wavelength in $\text{\AA}$ of the line in Balmer series that is associated with drop of electrons from the fourth orbit. $R = 109,676 \text{ cm}^{-1}$

(7 marks)

d) Electromagnetic radiation of wave length 242 nm is just sufficient to ionize the outermost electron of the sodium atom. Calculate the ionization energy of sodium atom ($c = 3 \times 10^8 \text{ ms}^{-1}$, $h = 6.625 \times 10^{-34} \text{ J.s.}$)

(5 marks)

(ii) Quantum number denoted by symbol l

(2 marks) ✓

d) Write the electronic configuration of the following species

(i) Mn^{2+} (ii) As^{3+} (iii) K^+ (iv) Mo

(4 marks) ✓

e) Draw the Lewis structure for the following compounds:

(i) IF_7

(2 marks) ✓

(ii) OsF_8

(2 marks)

(iii) BF_3

(2 marks) ✓

(iv) CH_4

(2 marks) ✓

QUESTION TWO (20 marks)

a) Below are general formulas of common organic compounds. Name these compounds and give their functional groups (4marks)

(i) $R-C\equiv C-H$

alkyne

(ii) $R-NH_2$

b) Explain the following giving appropriate reasons:

(i) Half-filled and completely-filled orbitals are more stable

(2 marks)

(ii) Fe^{2+} ion is less stable than Fe^{3+} ion

(2 marks)

c) In which group of the periodic table will be found:

(i) An element with atomic number 20

(1 mark)

(ii) An element having $2s^2 2p^2$ outer configuration

(1 mark)

(iv) An element with three unpaired p-electrons in the valence-shell

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d) Write the electronic configuration of the elements whose atomic numbers are 24, 29, 46, 56 and 78

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(5 marks)

QUESTION FOUR (20 marks)

a) Using two examples show that certain molecules deviate from the rule of eight and result in an expanded octet. (4 marks)

b) Give a comparison between Organic and inorganic compounds (4 marks) ✓

Volatility
covalent
larger
molecular
reaction
low MP and BP
Acid, Base, Salt
ionic
reactions
high MP and BP

c) Calculate the oxidation number of the following atoms:

(i) bromine in HOBr and KBrO₃ (4 marks) ✓

(ii) iodine in I₄O₇, ~~PI₅~~, KIO₃ (6 marks) ✓

3.5 +5

QUESTION FIVE (20 marks)

a) Arrange the following in increasing order of oxidation number

i) Manganese in MnCl₂, MnO₂, KMnO₄ (6 marks) ✓

b) Balance and complete the following equations by oxidation number method:

(i) $\text{Cr}_2\text{O}_7^{2-} + \text{I}^- \xrightarrow{\text{H}^+} \text{Cr}^{3+} + \text{I}_2 + \text{H}_2\text{O}$ (in acidic medium) (4 marks) ✓

(ii) $\text{AsO}_3^{2-} + \text{MnO}_4^- \rightarrow \text{AsO}_4^{6-} + \text{MnO}_2 + \text{H}_2\text{O}$ (in alkaline medium) (4 marks) ✓

c) Electrons emitted with zero velocity from a metal surface when it is exposed to radiations of 7200 Å. Calculate threshold frequency and work function of the metal (6 marks) ✓

ECU 100

Periodic Table of the Elements

Periodic Table of the Elements

1 1A 1A																	18 VIII 8A						
1 H Hydrogen 1.008	2 IIA 2A																	2 He Helium 4.003					
3 Li Lithium 6.941	4 Be Beryllium 9.012																	5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
11 Na Sodium 22.990	12 Mg Magnesium 24.305	3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8 VIII 8	9 VIII 8	10 VIII 8	11 IB 1B	12 IIB 2B	13 IIIA 3A	14 IVA 4A	15 VA 5A	16 VIA 6A	17 VIIA 7A	18 VIII 8A						
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.631	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798						
37 Rb Rubidium 84.46	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.905	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.411	49 In Indium 114.818	50 Sn Tin 118.710	51 Sb Antimony 121.757	52 Te Tellurium 127.6	53 I Iodine 126.905	54 Xe Xenon 131.29						
55 Cs Cesium 132.905	56 Ba Barium 137.327	57-71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.948	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.222	78 Pt Platinum 195.084	79 Au Gold 196.967	80 Hg Mercury 200.592	81 Tl Thallium 204.383	82 Pb Lead 207.2	83 Bi Bismuth 208.980	84 Po Polonium [209]	85 At Astatine [210]	86 Rn Radon [222]						
87 Fr Francium 223.021	88 Ra Radium 226.025	89-103	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [277]	109 Mt Meitnerium [268]	110 Ds Darmstadtium [271]	111 Rg Roentgenium [272]	112 Cn Copernicium [285]	113 Uut Ununtrium [284]	114 Fl Flerovium [289]	115 Uup Ununpentium [288]	116 Lv Livermorium [293]	117 Uus Ununseptium [294]	118 Uuo Ununoctium [294]						

Lanthanide Series

57 La Lanthanum 138.905	58 Ce Cerium 140.116	59 Pr Praseodymium 140.908	60 Nd Neodymium 144.242	61 Pm Promethium 144.913	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.925	66 Dy Dysprosium 162.500	67 Ho Holmium 164.930	68 Er Erbium 167.259	69 Tm Thulium 168.934	70 Yb Ytterbium 173.055	71 Lu Lutetium 174.967
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Actinide Series

89 Ac Actinium 227.028	90 Th Thorium 232.038	91 Pa Protactinium 231.036	92 U Uranium 238.029	93 Np Neptunium 237.048	94 Pu Plutonium 244.064	95 Am Americium 243.061	96 Cm Curium 247.070	97 Bk Berkelium 247.070	98 Cf Californium 251.083	99 Es Einsteinium [252]	100 Fm Fermium 257.103	101 Md Mendelevium 258.1	102 No Nobelium 259.101	103 Lr Lawrencium [260]
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